

Lesson 1 Computer Image Processing Fundamentals

OpenCV (Open Source Capture Vision) is a free computer vision library that can process images and Video is used to complete various tasks, such as displaying the signal input by the camera and allowing the robot to recognize objects in real life.

Although Python has its own image processing library PIL, its function is much weaker than OpenCV. OpenCV provides with a complete Python port, Python3.5 and python-opencv libraries have been integrated in the mirroring system we provide file. You can directly use this powerful computer vision library.

Next, I will briefly explain the basic knowledge of computer image processing, and the knowledge of OpenCV common functions.

We know that an image is composed of pixels, so how does the computer store and recognize different images? Each pixels can be represented by the R, G, and B values of the three primary colors ranging from 0-255. OpenCV uses a ternary for each pixel array representation, and then save this array to record all the information of the image. But it should be noted that the array of the three color channels of the RGB image recorded in OpenCV, the recording order will become BGR, so we often call the image is a BGR image.

For other images such as HSV and Lab standards, they are also stored in the form of a multivariate array. An OpenCV the image is a two-dimensional or three-dimensional array of type .array, and an 8-bit grayscale image (only black and white images) is a two-dimensional array, a 24-bit BGR image is a

three-dimensional array. For example, for a BRG image, `image[0,0,0]` the first value represents the Y coordinate or row of the pixel, and 0 represents the top; the second value represents the x coordinate or column of the pixel, and 0 represents the far left; the third value represents the color channel.

These arrays of recorded images can be accessed separately like ordinary Python arrays to obtain a certain color separately the value of the channel, or the image of a certain area of the image.